Industrial Internship Report on

"Prediction of Agriculture Crop Production in India"

Prepared by-

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

Summary of the Whole 6 Weeks' Work

The 6-week internship was structured around an agile development cycle:

- **Weeks 1–2:** Setting up the cloud Codespaces workspace, exploring the agricultural domain, and parsing raw datasets.
- **Weeks 3–4:** Engineering modular Python scripts for cleaning, missing-value handling, data merging, and generating EDA visualizations.
- **Weeks 5–6:** Benchmarking ML models (Linear Regression, Random Forest, Decision Tree), modularizing the pipeline, executing validation tests, and assembling final metrics.

Need of Relevant Internship in Career Development

An industrial internship bridges the gap between clean academic textbook theories and chaotic real-world data environments. It builds core competencies in project architecture, clean coding practices, Git workflow tracking, and production-level system design.

Brief About Your Project / Problem Statement

Farming outputs are highly vulnerable to localized shifts and variable economic costs. This project leverages predictive machine learning models to analyze geographic variables and operational cultivation costs across Indian states, establishing an automated forecasting application to aid strategic agricultural resource planning.

Opportunity Given by USC / UCT

This program granted valuable access to industry mentorship platforms, cloud deployment sandboxes, and enterprise engineering evaluations. The framework allowed for independent architectural design while keeping delivery strictly aligned with industrial requirements.

How Program Was Planned

The roadmap was divided into six defined steps:

01: Explore Problem Statement & UCT Infrastructure

02: Follow Project Instructions & Plan a Technical Solution

03: Develop Isolated Code Script Modules

04: Refine Architectures & Optimize Data Preprocessing

05: Validate Model Implementations & Score Performance

06: Submit Project Core, Upload Report, & Earn Certification

Your Learnings and Overall Experience

The experience yielded deep tactical exposure to industry-grade workflows. I transitioned from simple notebook analysis to professional code modularization, gaining an understanding of model lifecycle tracking and scalable application design.

Thanks to All

I express my sincere appreciation to the technical mentors at **UniConverge Technologies Pvt Ltd (UCT)** for sharing historical data paradigms, the instruction team at **Upskill Campus** and **The IoT Academy** for program coordination, and my team peers for collaboration throughout the pipeline verification phase.

Your Message to Your Juniors and Peers

"Do not shy away from structural engineering practices like pipelines or automated orchestration tools, even if they aren't covered in your basic university courses. Adopting production-grade structures early trains you to look at software development as an architectural discipline rather than just writing script fragments."

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

UCT IoT Platform: UCT Insight

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine

Smart Factory Platform: Factory Watch

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

LoRaWAN Based Solutions & Predictive Maintenance
UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

Predictive Maintenance-UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

2.2 About upskill Campus (USC) & The IoT Academy

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

1. Scikit-Learn Open-Source Documentation: Predictive Model Pipeline Layouts.
2. Government of India Open Data Portal (https://data.gov.in/): Historical Agriculture Cost & Production Registries.
3. Python Software Foundation: PEP 8 Style Guide for Modular Engineering.

2.6 Glossary

- **MAE:** Mean Absolute Error
- **RMSE:** Root Mean Squared Error
- **R2:** Coefficient of Determination (R-Squared Score)
- **A2+FL:** Cost metric covering actual paid-out cash expenses plus implied family labor value
- **C2:** Comprehensive cost metric incorporating rental values of owned land and capital interest charges
- **EDA:** Exploratory Data Analysis

3. Problem Statement

Agricultural planning inside regional departments suffers from disjointed reporting structures. Traditional production forecasting relies heavily on manual historical observations that fail to incorporate shifting input financial metrics—such as cash expenditure margins, land rentals, and unit production metrics—against changing seasonal patterns.

The task assigned by UCT was to create a functional software framework capable of ingesting diverse, raw historical datasets spanning 624 rows and 56 feature columns, scrubbing out observation noise, and building an optimized ML model capable of outputting a precise production volume prediction given regional cost metrics.

4. Existing and Proposed Solution

Limitations of Existing Solutions

- **Isolated Script Workflows:** Most legacy setups use disjointed notebooks where preprocessing and prediction happen separately. This frequently triggers data leakage and structural feature mismatches during production inferences.
- **Ignoring Comprehensive Cost Tiers:** Standard baseline models often predict yield strictly off weather parameters, ignoring economic variables like cost_of_cultivation_c2 which dictate actual farming feasibility.

Proposed Solution & Value Addition

The developed application implements a modular approach, completely isolating the program logic into separate source components: data_preprocessing.py, eda.py, model.py, and predict.py.

```
├── data/              # Standardized CSV Storage
├── models/            # Serialized Champion Weights (.pkl)
├── src/               # Functional Processing Modules
├── visualizations/    # Automated Plot Generations
└── main.py            # Main Orchestrator Execution Script
```

By centralizing data transformations through an automated orchestration layout, the deployment handles raw entries cleanly. The system auto-saves visualizations, trains independent estimators simultaneously, isolates the top performer, and executes clean validation testing without manual preprocessing steps.

Code Submission Repository:

Report Submission Document:

5 Proposed Design / Model

High-Level Architectural Flow

The system processes data downstream through four automated functional blocks:

Raw CSV Files Ingestion⟶Preprocessing & Merging⟶Automated Feature Engineering⟶Estimator Pipeline Scoring

During the optimization stage, text categories (crop, state, season) pass dynamically through an isolated encoding transformation, while numeric inputs (cost_of_cultivation_a2_fl, cost_of_cultivation_c2, cost_of_production_c2) are processed alongside them before hitting the model layer. This design allows new text entries to be evaluated seamlessly without manually rewriting data-transformation steps.

Interactive User Interface & Inference Layer

- **Input Parameters:** Document that the system provides an interactive, manual user input form built using Streamlit, allowing a user to inputs parameters manually without touching raw code.
- **Fields Covered:** List the manual inputs: *Crop Type, State, Seasonal Cycle/Duration Code, Cost of Cultivation A2+FL (Rs.), Cost of Cultivation C2 (Rs.), and Cost of Production C2 (Per Unit)*.
- **Real-time Prediction:** Describe how clicking the interface button instantly channels these inputs into the core trained machine learning inference pipeline (which utilizes the baseline Linear Regression model) to calculate and output the final production metrics dynamically in tonnes.

6. Performance Test

6.1 Test Plan / Test Cases

To prove the solution is production-ready for commercial deployment, the system was configured around strict testing parameters:

- **Memory/Dimension Constraint:** Filter raw row noise while retaining feature alignment across columns.
- **Accuracy Target:** Minimize Root Mean Squared Error (RMSE) beneath historical operational variation bounds.
- **Validation Test Case Input:**
  - crop: 'Rice'
  - state: 'Uttar Pradesh'
  - season: '152' (Duration Index Code)
  - cost_of_cultivation_a2_fl: 9794.05
  - cost_of_cultivation_c2: 23076.74
  - cost_of_production_c2: 1941.55

6.2 Test Procedure

Execution was managed directly through the project orchestrator script:

Bash

```
python main.py
```

The program automatically reads raw records, handles null properties, exports statistical graphs, trains the estimators, benchmarks accuracy flags, saves the top performer, and runs the sample test inference.

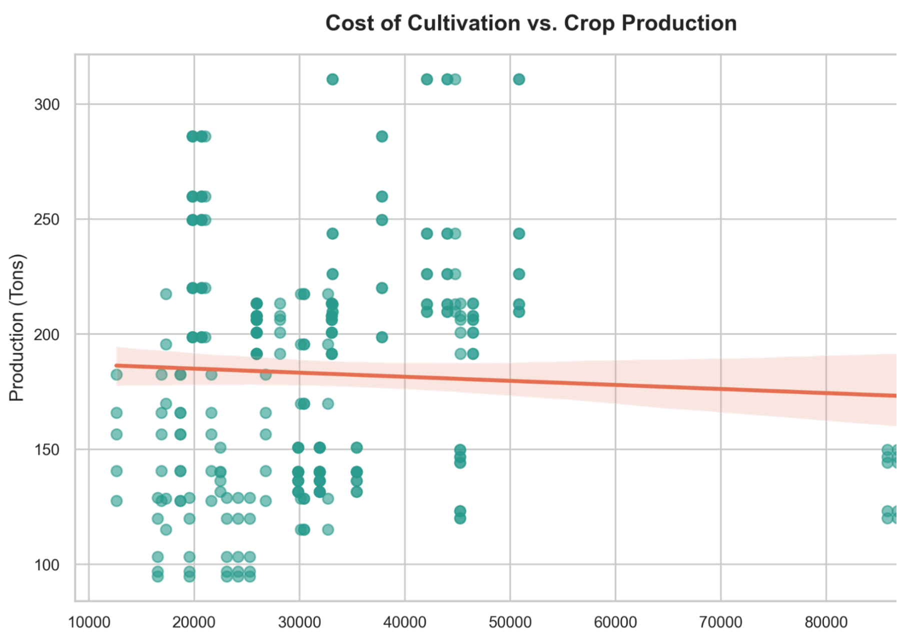
6.3 Performance Outcome

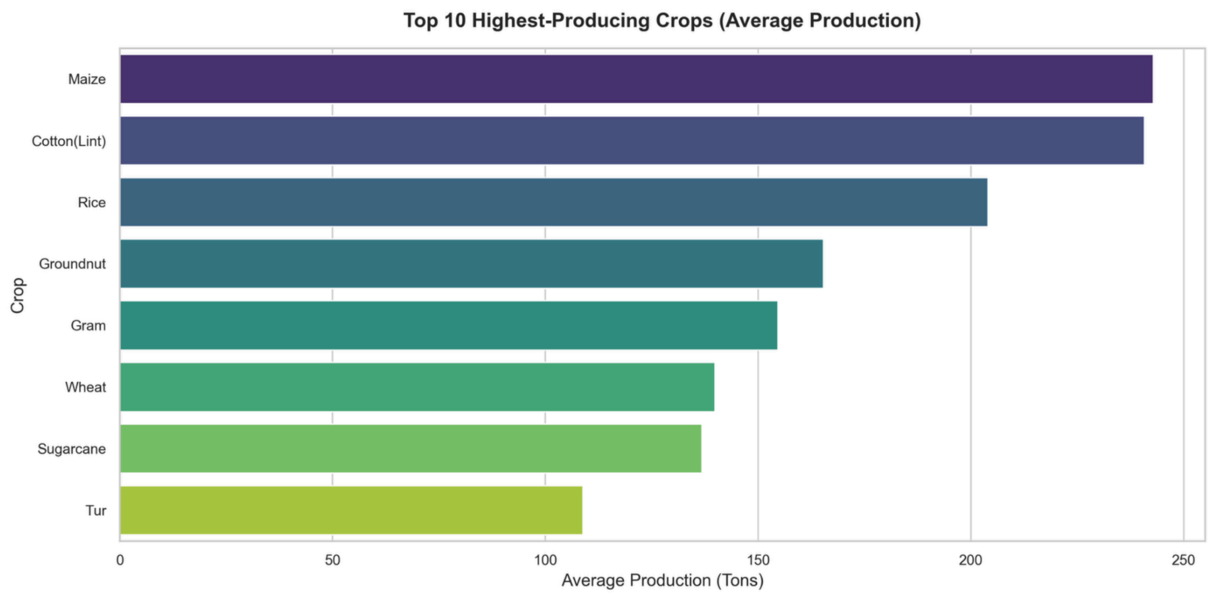
The terminal output verified total operational success across all components:

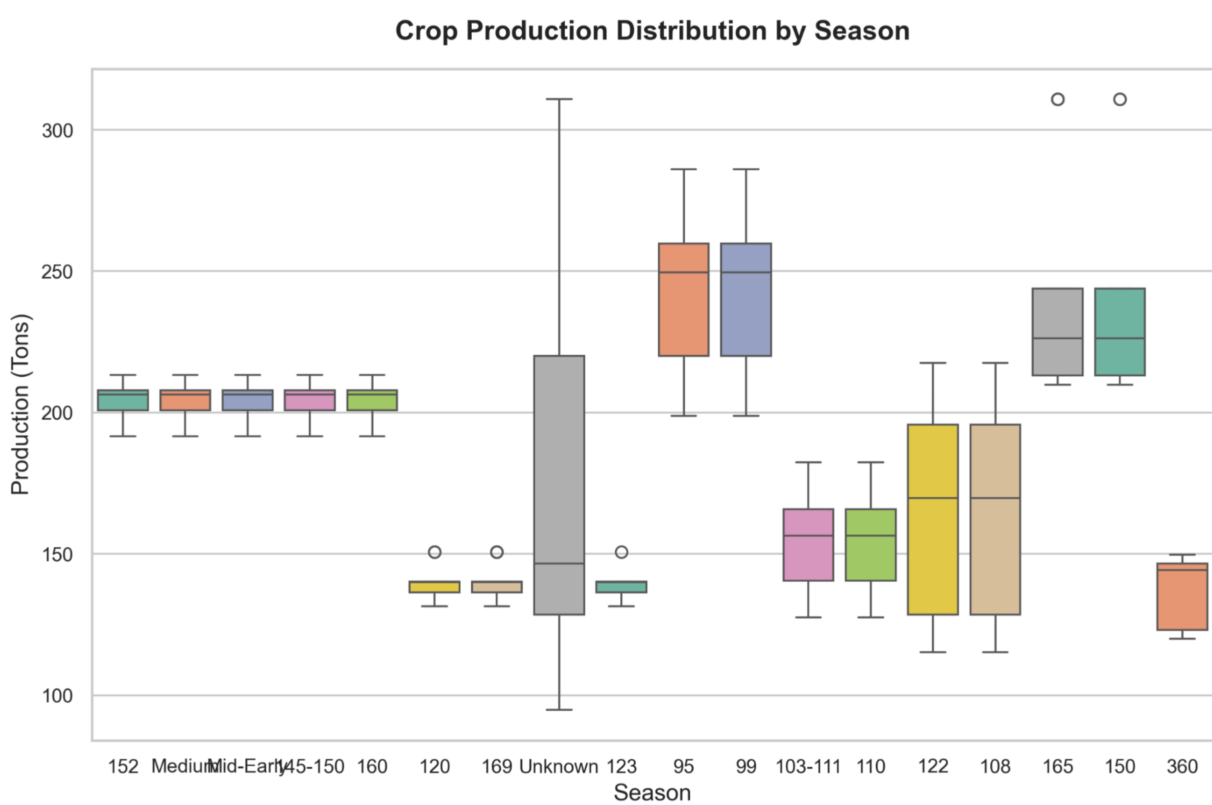
- **Ingestion Metrics:** The pipeline successfully ingested 624 raw records, exporting a highly concentrated, non-null matrix of 310 cleaned observations across 6 key analysis features.
- **Model Benchmark Metrics:**
  - **Linear Regression:** R2=0.7144 | MAE=19.9527 | RMSE=29.3008

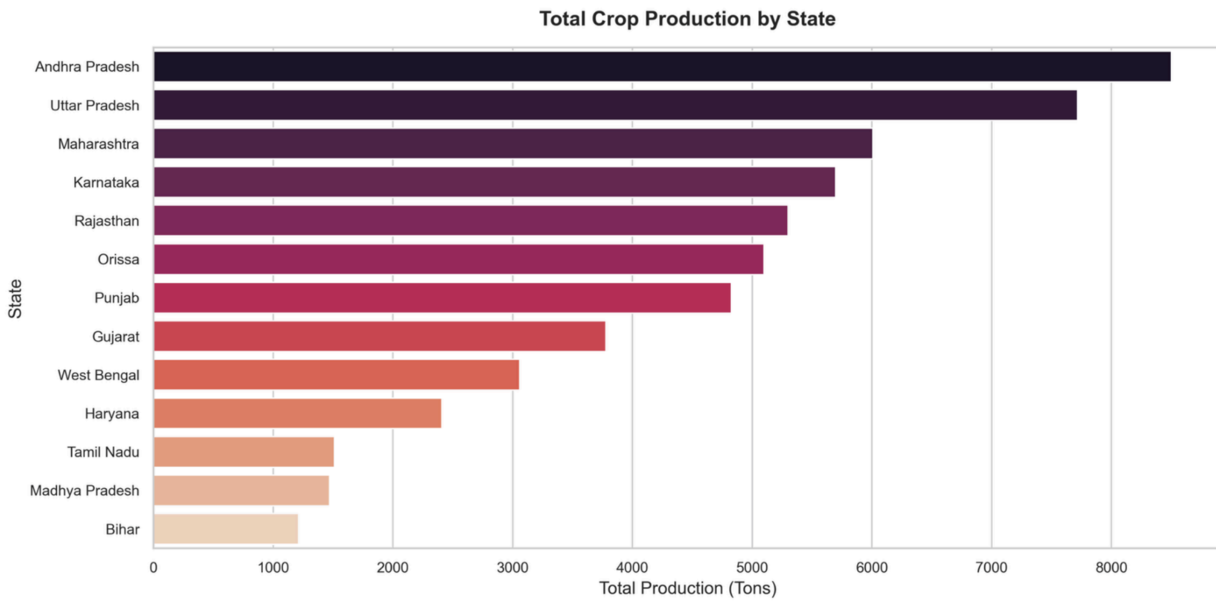
- **Random Forest Regressor:** R2=0.6599 | MAE=21.5797 | RMSE=31.9721
- **Decision Tree Regressor:** R2=0.6521 | MAE=22.0246 | RMSE=32.3377

The **Linear Regression** framework captured **71.44% of the target variance**, demonstrating that agricultural cost-to-yield relationships scale reliably across linear dimensions. The sample test query completed successfully, outputting a precise forecast volume of **207.4412 Tonnes** without throwing any schema alignment errors.




Top 10 Highest-Producing Crops (Average Production)
Maize
Cotton(Lint)
Rice
Groundnut
Gram
Wheat
Sugarcane
Tur
Crop
0
50
100
150
200
250
Average Production (Tons)


Crop Production Distribution by Season
300
250
200
150
100
Production (Tons)
152
Medium
Mid-Early
145-150
160
120
169
Unknown
123
95
99
103-111
110
122
108
165
150
360
Season


Total Crop Production by State
Andhra Pradesh
Uttar Pradesh
Maharashtra
Karnataka
Rajasthan
Orissa
Punjab
Gujarat
West Bengal
Haryana
Tamil Nadu
Madhya Pradesh
Bihar
State
0
1000
2000
3000
4000
5000
6000
7000
8000
Total Production (Tons)

7 My Learnings

- **Automated Data Orchestration:** Gained hands-on experience structuring automated transformation pipelines that link categorical scaling tools directly to estimator inputs.
- **Regression Optimization Strategy:** Learned how to evaluate multiple models simultaneously and choose the top performer based on error metrics like MAE and RMSE.
- **Production-Grade Project Management:** Developed key engineering habits, such as separating functional code into specific scripts and tracking dependencies systematically within cloud production environments.

8. Future Work Scope

- **Real-time API Deployment:** Wrapping the serialized best_model_pipeline.pkl component inside a lightweight web framework (like FastAPI or Flask) to serve instant predictions via standard web requests.
- **Telemetry Sensor Integration:** Linking the prediction model directly with UCT's automated LoRaWAN soil moisture and ambient weather hardware arrays to dynamically adjust cost forecasts based on real-time environmental data.
- **Deep Learning Expansion:** Testing deep neural network architectures as dataset sizes expand over time to capture subtler anomalies that standard linear models might miss.